

① Does there exist a bijective map $\pi: \mathbb{N} \rightarrow \mathbb{N}$ such that $\sum_{n=1}^{\infty} \frac{\pi(n)}{n^2} < \infty$?

Assume there exists a bijection $\pi: \mathbb{N} \rightarrow \mathbb{N}$ s.t. the series $\sum_{n=1}^{\infty} \frac{\pi(n)}{n^2}$ converges. $\Rightarrow (S_N)$ is bounded

Let $S_N = \sum_{n=1}^N \frac{\pi(n)}{n^2}$

Suppose that for some indices $i < j$, we have $\pi(i) > \pi(j)$.
 $\Rightarrow$ the permutation is not ordered increasingly at those 2 positions
 (since π is a bijection, we can treat it as a permutation)

If we swap them

Before: $\frac{\pi(i)}{i^2} + \frac{\pi(j)}{j^2}$		Before - After
After: $\frac{\pi(j)}{i^2} + \frac{\pi(i)}{j^2}$		$\frac{\pi(i)}{i^2} + \frac{\pi(j)}{j^2} - \frac{\pi(j)}{i^2} - \frac{\pi(i)}{j^2} =$ $= \pi(i) \left(\frac{1}{i^2} - \frac{1}{j^2} \right) + \pi(j) \left(\frac{1}{j^2} - \frac{1}{i^2} \right) =$ $= \underbrace{\left(\frac{1}{i^2} - \frac{1}{j^2} \right)}_{>0} \underbrace{\left(\pi(i) - \pi(j) \right)}_{>0} > 0$

$\Rightarrow$ swapping such a pair decreases the total sum

By applying this repeatedly, ~~we~~ eventually, there are no inversions left $\Rightarrow \pi(1) < \pi(2) < \dots < \pi(N)$

But the only permutation with no inversions is the identity permutation.

$\Rightarrow \pi(i) = i$ gives the smallest possible value of the partial sum

$$\Rightarrow \sum_{n=1}^N \frac{\pi(n)}{n^2} \geq \sum_{n=1}^N \frac{n}{n^2} = \sum_{n=1}^N \frac{1}{n} \rightarrow \infty$$

$\hookrightarrow$ the harmonic series

$\Rightarrow S(\pi) = \lim_{N \rightarrow \infty} S_N(\pi) = \infty$, which contradicts the initial assumption

$\Rightarrow$ No, there isn't a bijective map $\pi: \mathbb{N} \rightarrow \mathbb{N}$ such that $\sum_{n \geq 1} \frac{\pi(n)}{n^2} < \infty$

② Let (x_n) be a decreasing sequence of positive numbers. Show that if $\sum_{n \geq 1} x_n$ converges, then $\lim_{n \rightarrow \infty} n x_n = 0$.

~~Cauchy Condensation Test~~

~~(x_n) decreasing sequence with $x_n > 0 \Rightarrow$
 $\Rightarrow$ The series $\sum_{n=1}^{\infty} x_n$ and $\sum_{n=0}^{\infty} 2^n x_{2^n}$ have the same nature~~

Let $S_N = \sum_{k=1}^N x_k$ and $T_m = \sum_{k=m+1}^{\infty} x_k$

the rest of the terms that were not in S_N

$\sum_{n \geq 1} x_n$ converges $\Rightarrow$ as $m \rightarrow \infty$, $T_m \rightarrow 0$ (**)

Let $m \in \mathbb{N}$, $m \geq 1$

Let $m = \lfloor n/2 \rfloor \Rightarrow m \leq \frac{n}{2} \Rightarrow n - m \geq \frac{n}{2}$ (*)

(x_n) is decreasing and positive $\Rightarrow$

$\Rightarrow$ for all k with $m+1 \leq k \leq n$, we have $x_k \geq x_m$

There are $n-m$ such integers $(k) \Rightarrow$

$$\Rightarrow \sum_{k=m+1}^n x_k \geq (n-m) x_m \stackrel{(*)}{\geq} \frac{n}{2} x_m$$

$$\text{But } \sum_{k=m+1}^n x_k \leq \underbrace{\sum_{k=m+1}^{\infty} x_k}_{T_m}$$

$$\Rightarrow \frac{n}{2} x_m \leq T_m \quad | \cdot 2$$

$$n x_m \leq 2 \cdot T_m$$

$$\text{If } n \rightarrow \infty \Rightarrow \frac{n}{2} \rightarrow \infty \Rightarrow T_m \rightarrow 0 \quad (**)$$

$$\Rightarrow \limsup_{n \rightarrow \infty} n x_n \leq 0$$

$$\text{But } x_n > 0 \Rightarrow n x_n \geq 0, \quad \forall n \in \mathbb{N}$$

$$\Rightarrow 0 \leq \liminf_{n \rightarrow \infty} n x_n \leq \limsup_{n \rightarrow \infty} n x_n \leq 0$$

$$\Rightarrow \lim_{n \rightarrow \infty} n x_n = 0.$$

③ Start with the interval $[0, 1]$ and remove the middle third $(1/3, 2/3)$; then remove the middle third of each of the remaining intervals, and so on. Denote by C set of points that are left - this is known as the Cantor set.

(•) What is the total length of the removed intervals?

Step 1: remove the middle open interval $(1/3, 2/3)$ of length $1/3 = \frac{1}{3^1} = \frac{2^0}{3^1}$

Step 2: remove the 2 middle-thirds (one from each remaining interval), each of length $\frac{1}{3^2} \Rightarrow$ total

removed (at step 2) $2 \cdot \frac{1}{3^2} = \frac{2}{3^2} = \frac{2^1}{3^2}$

⋮

Step k : remove 2^{k-1} intervals, each of length $\frac{1}{3^k} \Rightarrow$
 $\Rightarrow$ total removed at step $k = \frac{2^{k-1}}{3^k}$

$\Rightarrow$ Total length removed in all stages

$$\sum_{k=1}^{\infty} \frac{2^{k-1}}{3^k} = \sum_{k=1}^{\infty} \frac{1}{3} \cdot \frac{2^{k-1}}{3^{k-1}} = \frac{1}{3} \sum_{k=1}^{\infty} \left(\frac{2}{3}\right)^{k-1} = \frac{1}{3} \cdot \frac{1}{1 - \frac{2}{3}} = \frac{1}{3} \cdot \frac{1}{\frac{1}{3}} = \frac{1}{3} \cdot 3 = \boxed{1}$$

(••) Prove that $x \in C$ iff $x = \sum_{n=1}^{\infty} a_n 3^{-n}$, with $a_n \in \{0, 2\}$.
 In other words, $x \in C$ iff it can be written in base 3 using only 0's and 2's.

$\Rightarrow$ If $x \in C \Rightarrow$ it can be written only with 0's and 2's in base 3.

$x = 0.a_1a_2a_3\dots$

At step 1: the remaining set is $[0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$

So any point is either in the left third ($a_1 = 1$) or in the right third ($a_1 = 2$)

At step 2: each remaining third is split into 2 subintervals, x is found in one of them $\Rightarrow a_2=0$ & $a_2=2$
:

At step m : x lies in one of 2^m closed intervals of length 3^{-m}

All these choices determine a sequence $(a_m)_{m \geq 1}$ with $a_m \in \{0, 2\}$

$$\Rightarrow x = \sum_{m=1}^{\infty} \frac{a_m}{3^m}$$

$$\Leftarrow \text{If } x = \sum_{m=1}^{\infty} a_m 3^{-m} \text{ with } a_m \in \{0, 2\} \Rightarrow x \in C$$

a_1 = choose left / right third at stage 1

a_2 = choose left / right subthird at stage 2

...

At each stage m , the expression $\sum_{k=1}^m a_k 3^{-k}$ is the left endpoint of the closed interval of length 3^{-m} in which x lies.

But $a_k \in \{0, 2\} \Rightarrow$ that interval is one of the "good" ones at stage m .

$\Rightarrow x$ is in every stage's "good" set $\Rightarrow$

$\Rightarrow x$ is in the intersection of all sets $\rightarrow$
 $\Rightarrow x \in C$

(...) By constructing a surjective function from C to $[0, 1]$, prove that C is uncountable.

From (...), we know that every $x \in C$ can be written in base 3 using only 0's and 2's

$$x = \sum_{m=1}^{\infty} \frac{a_m}{3^m}, \quad a_m \in \{0, 2\}$$

$$\text{Let } b_m = \frac{a_m}{2} \Rightarrow b_m \in \{0, 1\}$$

$$\Rightarrow x = \sum_{m=1}^{\infty} \frac{2 b_m}{3^m}, \quad b_m \in \{0, 1\}$$

$$f: \mathbb{C} \rightarrow [0, 1]$$

$f(x)$ will be the binary representation of a number in $[0, 1]$

$$\Rightarrow f(x) = \sum_{m=1}^{\infty} \frac{b_m}{2^m}$$

$$b_m \in \{0, 1\} \Rightarrow f(x) \in [0, 1]$$

Prove that f is surjective:

Take $y \in [0, 1]$. Its binary representation will be:

$$y = \sum_{m=1}^{\infty} \frac{b_m}{2^m}, \quad b_m \in \{0, 1\}$$

$$a_m = 2 b_m \in \{0, 2\}$$

$$\Rightarrow x = \sum_{m=1}^{\infty} \frac{a_m}{3^m} \in \mathbb{C}$$

$$f(x) = \sum_{m=1}^{\infty} \frac{b_m}{2^m} = y$$

$\Rightarrow$ every $y \in [0, 1]$ is the image of an element in $\mathbb{C} \Rightarrow f$ is surjective

$[0, 1]$ is uncountable

f is a surjection from $\mathbb{C}$ onto $[0, 1]$, $\Rightarrow \mathbb{C}$ is also uncountable

$\therefore \mathbb{C}$ has at least as many elements as $[0, 1]$